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WAIT WATER COLLECTION SYSTEM

Abstract

The wait water collection system provided herein may be used to reducing the amount of water wasted while a shower, for example, warms up. The wait water collection system may include a hot water input carrying hot water to the system. The hot water may or may not be “hot” when delivered by the hot water input and may need to “warm up.” The wait water collection system may include a thermal diverter on a water line where the hot water travels, which may adjust based on whether the temperature of the hot water meets a temperature threshold. If the water does not meet the temperature threshold, then the hot water may be delivered to an expansion vessel. If the water meets or exceeds the temperature threshold, then the hot water may be delivered to a mixer valve for providing the water to a user.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority to U.S. Provisional Patent Application No. 63/559,263, filed Feb. 29, 2024, entitled, “WAIT-WATER COLLECTION SYSTEM,” and to U.S. Provisional Patent Application No. 63/559,288, filed Feb. 29, 2024, entitled, “SYSTEM AND METHODS FOR MANAGEMENT OF WATER,” the entire contents of which are hereby incorporated by reference.

BACKGROUND

[0002] At times, consumers may turn on a faucet (e.g., hot water valve) and wait for the water to reach a certain temperature before using the water. For example, consumers may wait to enter a shower until the water has reached a desired temperature. Water that has flowed through the system while the user waits for the temperature to warm up may enter the drain and septic system without use. The amount of this water, which may be referred to as “wait water,” may be considerable. It may be desirable for an automated system to collect the wait water for reuse in another device (e.g., toilet, cold water tap, or irrigation).

[0003] Further, stagnant water tends to occupy pipes that are not in use or are between uses. Generally, stagnant water is discarded when the pipe is next used. This may result in wasted water, higher utility payments, or other undesirable outcomes. Therefore, there is a need for a pipe that limits or eliminates the presence of stagnant water when the pipe is not in use.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0004] FIG. 1 is a schematic view of an example wait water collection system.

[0005] FIG. 2 is a perspective view of an example expansion vessel.

[0006] FIG. 3 is a perspective view of an example expansion vessel.

[0007] FIG. 4 is a front elevation view of an example drain gate.

[0008] FIG. 5 is a cross-sectional view of an example drain gate in a first stage of operation.

[0009] FIG. 6 is a cross-sectional view of an example drain gate in a second stage of operation.

[0010] FIG. 7 is a perspective view of an example venturi.

[0011] FIG. 8 is a cross-sectional view of an example venturi.

[0012] FIG. 9 is a schematic view of an example wait water collection system.

[0013] FIG. 10 is a schematic view of an example wait water collection system.

[0014] FIG. 11 is a perspective view of an example thermostatic diverter.

[0015] FIG. 12 is a side cross-sectional of an example thermostatic diverter.

[0016] FIG. 13 is a schematic view of an example wait water collection system.

[0017] FIG. 14 is a schematic view of an example wait water collection system.

[0018] FIG. 15 is a front elevation view of an example toilet with an example storage tank.

[0019] FIG. 16 is a schematic view of an example wait water collection system.

[0020] FIG. 17 is a schematic view of an example wait water collection system.

[0021] FIG. 18 is a schematic view of an example wait water collection system.

[0022] FIG. 19 is a schematic view of an example wait water collection system.

[0023] FIG. 20 is a schematic view of an example wait water collection system.

[0024] FIG. 21 is a schematic view of an example wait water collection system.

[0025] FIG. **22** is a schematic view of an example wait water collection system.
[0026] FIG. **23** is a schematic view of an example wait water collection system.
[0027] FIG. **24** is a schematic view of an example wait water collection system.
[0028] FIG. **25** is a schematic view of an example wait water collection system.
[0029] FIG. **26** is a perspective view of an example pipe with a flexible membrane.
[0030] FIG. **27** is a perspective cross-sectional view of an example pipe in a first stage of operation.

[0031] FIG. **28** is a perspective cross-sectional view of an example pipe in a second stage of operation.

[0032] While the disclosure is susceptible to various modifications and alternative forms, a specific embodiment thereof is shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description presented herein are not intended to limit the disclosure to the particular embodiment disclosed, but to the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

DETAILED DESCRIPTION

[0033] The invention will now be described with reference to the drawing figures, in which like reference numerals refer to like parts throughout. For purposes of clarity in illustrating the characteristics of the present invention, proportional relationships of the elements have not necessarily been maintained in the drawing figures.

[0034] As stated above, it may be desirable to have a wait water collection system to reduce water usage and water waste. A wait water collection system **1** may be a “one-box unit,” which may be installed to aid in preventing the waste of wait water. The wait water collection system **1** may include several components, as discussed herein, which may operate to aid in preventing the waste of wait water. The wait water collection system **1** may operate without additional electrical power, and the wait water collection system **1** may be operated using water pressure alone.

[0035] Hot water may enter a wait water collection system **1** via a hot water input **5**. The hot water may travel via a first water line **10** to a thermal diverter **15**, which may include a valve that adjusts based on a temperature. If the water meets or exceeds a threshold temperature, the thermal diverter **15** may cause the water to flow to a mixer valve **20** (which may have a hot water inlet **26** and a cold water inlet **30**) via a second water line **35**. The mixer valve **20** may be any mixer valve now known or hereafter developed which may be used to adjust a temperature of water. A user may select a temperature setting, which may cause the wait water collection system **1** (and components therein) to adjust to output water at the selected temperature. The water mixed via the mixer valve **20** may exit the wait water collection system **1** via an outlet **40** (e.g., a showerhead, handset, faucet, etc.).

[0036] If the water is below a threshold temperature (which may be predetermined), the valve within the thermal diverter **15** may cause the water to flow through a third water line **45** into an expansion vessel **50** via an expansion vessel port **55**. The expansion vessel **50** may be any suitable capacity. As one example, the expansion vessel **50** may have a capacity of 8 liters or 2.11 gallons; as another example, the expansion vessel **50** may have a capacity of 6 to 16 liters. The expansion vessel **50** may include an air inlet **60** which may be used to adjust the air pressure within the expansion vessel **50**. The air pressure may be adjusted, for example, depending on the water pressure being delivered to the system. A user may access the air inlet **60** in order to alter the air pressure in the expansion vessel **50**. Even when the expansion vessel **50** is not easily accessible by a user (e.g., behind a wall), the air inlet **60** may be attached to an air filler tube, which the user may access to alter the air pressure in the expansion vessel **50**. The air filler tube may include a standard tyre valve, which may be connected to a standard pump and air pressure gauge to change the air pressure.

[0037] Cold water may enter the wait water collection system **1** via a cold water input **65**. The cold water may flow through a fourth water line **70** and may pass through a venturi **75**. The venturi **75**

may be used to draw water out from the expansion vessel **50**. The water from the expansion vessel **50** may exit the expansion vessel **50** via the expansion vessel port **55** and may flow to the venturi **75** via a fifth water line **80**. The water from the expansion vessel **50** may mix with the water from the cold water input **65** and may be directed into the mixer valve **20** via a sixth water line **85**.

[0038] A pressure sensor **90** may be positioned on the second water line **35**. The pressure sensor **90** may determine a pressure of the water passing through the second water line **35**, and the pressure detected by the pressure sensor **90** may be sent to each of an inlet gate valve **95** and a drain gate valve **100**. The pressure may be sent via a pilot tube **105** connecting the pressure sensor **90** to each of the inlet gate valve **95** and the drain gate valve **100**. When the water pressure is low, the inlet gate valve **95** may be opened and the drain gate valve **100** may be closed. When the water pressure is high, the inlet gate valve **95** may be closed and the drain gate valve **100** may be opened. When the inlet gate valve **95** is opened, hot water may flow in via the hot water input **5** to the thermal diverter **15**. When the drain gate valve **100** is opened, water may flow out of the expansion vessel **50** through the outlet **40** via a seventh water line **110**.

[0039] As a first example, when the wait water collection system **1** is in a “static” state (i.e., a previous user has switched off the mixer valve **20** when hot water was flowing), the pressure sensor **90** may detect a high pressure. As such, the inlet gate valve **95** may close, and the drain gate valve **100** may open such that water from the expansion vessel **50** flows out via the outlet **40**. As a second example, when a user switches on the shower with a hot temperature selected, the pressure sensor **90** may detect a low pressure. As such, the inlet gate valve **95** may open, and the drain gate valve **100** may close such that water from the expansion vessel **50** flows into the venturi **75** and mixes with the cold water. As a third example, when cold water flows into the expansion vessel **50**, the pressure sensor **90** may detect a low pressure. As such, the inlet gate valve **95** may open, and the drain gate valve **100** may close such that water from the expansion vessel **50** flows into the venturi **75** and mixes with the cold water.

[0040] As a fourth example, when the hot leg arrives, the pressure sensor **90** may detect a low (but rising) pressure. As such, the inlet gate valve **95** may open, and the drain gate valve **100** may close such that water from the expansion vessel **50** flows into the venturi **75** and mixes with the cold water. As a fifth example, when the hot leg is flowing, the pressure sensor **90** may detect a medium (but not high) pressure. As such, the inlet gate valve **95** may open, and the drain gate valve **100** may close such that water from the expansion vessel **50** flows into the venturi **75** and mixes with the cold water. As a sixth example, when a user switches off the unit, the pressure sensor **90** may detect a high pressure. As such, the inlet gate valve **95** may close, and the drain gate valve **100** may open such that water from the expansion vessel **50** flows out via the outlet **40**. At the end of the showering period, the expansion vessel **50** may be emptied. The foregoing operational examples are provided for exemplary purposes only and should not be construed as limited.

[0041] The pressure sensor **90** connectivity to the inlet gate valve **95** and the drain gate valve **100** may be useful in compensating for failures of the wait water collection system **1**. For example, if a user switches off the mixer valve **20** before the hot leg arrives, the pressure sensor **90** may detect a low pressure, and as such, the drain gate valve **100** may be closed. Water may continue to flow into the expansion vessel **50** even though the mixer valve has been switched off. Once the hot leg arrives, the thermal diverter **15** may send the hot water to the mixer valve **20**. The pressure sensor **90** may detect a high pressure, and as such, the drain gate valve **100** is opened. The expansion vessel **50** may then empty, and although some water may be wasted in the process, the wait water collection system **1** may be ready for a next user.

[0042] In the circumstance where the hot leg does not arrive (e.g., due to a fault of the water heater), the thermal diverter **15** may send ambient water to the expansion vessel **50**. Once the expansion vessel **50** is filled, the wait water collection system **1** may lock. The wait water collection system **1** may be primed for a new user once the water heater fault is fixed. In order to be primed for a new user, a user may switch the mixer valve **20** “on” with a set temperature of cold,

such that the water from the expansion vessel **50** is drained through the outlet **40** via the venturi **75** and mixer valve **20**.

[0043] When in such a configuration, the expansion vessel **50** may be oriented vertically (as illustrated in FIG. 2) or horizontally (as illustrated in FIG. 3). In either orientation, a base **115** of the expansion vessel **50** may be slanted, such that water in the expansion vessel **50** drains toward the expansion vessel port **55**. The expansion vessel port **55** may be positioned at an edge **120** of the expansion vessel **50** where the base **115** slants downward. Such a configuration may aid in ensuring that water does not collect in the expansion vessel **50** and that water is successfully drained out of the expansion vessel **50**. It should be understood that any other suitable orientation or positioning may be achieved and that FIGS. 2 and 3 should not be construed as limiting.

[0044] Turning now to FIG. 4, the drain gate valve **100** as discussed herein may adjust based on the detected pressure by the pressure sensor **90**. The drain gate valve **100** may be positioned on the seventh water line **110** near or adjacent to the second water line **35**. Turning to FIG. 5, the drain gate valve **100** may include a rocker plunger **125**, which may be operated based on the pressure of water flowing in the second water line **35**. When high pressure is detected, a diaphragm **130** within the drain gate valve **100** may press in a downward direction on a first end **135** of the rocker plunger **125**. This may cause a second end **140** of the rocker plunger **125** to lift, which may allow the drain gate valve **100** to open, causing the water in the expansion vessel **50** to drain.

[0045] As illustrated in FIG. 6, when low pressure is detected, the diaphragm **130** may remain in place, causing the second end **140** of the rocker plunger **125** to drop. This may close the drain gate valve **100**, causing the water from the expansion vessel **50** to flow through the venturi **75** and mix with the cold water.

[0046] The venturi **75** as discussed herein may be used to draw water out of the expansion vessel **50**. The venturi **75** may be any venturi now known or hereafter developed, and as one example, the venturi **75** may be the venturi as disclosed in U.S. patent application Ser. No. 18/947,734, filed Nov. 14, 2024, entitled, "POD VENTURI," the entire contents of which are hereby incorporated by reference. The venturi **75** may include multiple venturis for installation, as illustrated in FIG. 7. As one example, the venturi **75** may include a high pressure venturi **145** and a low pressure venturi **150**, one of which may be installed for use.

[0047] When the wait water collection system **1** is being installed, an installer may determine whether the high pressure venturi **145** or the low pressure venturi **150** is suitable for the situation. If the installer determines that the water flowing in the system is at a high pressure, for example, the high pressure venturi **145** may be installed inline with the applicable water line. The venturi **75** may include an arrow **155** indicating which direction water should flow when the venturi **75** is installed, which may be useful for installation purposes. The venturi **75** may include an expansion vessel input **160** which may input water from the expansion vessel **50** into the venturi **75**. When installed, the expansion vessel input **160** may be connected to the fifth water line **80**. Each end of the venturi **75** may include surfaces where o-ring seals may be installed to aid in ensuring a tight seal with the water lines.

[0048] Turning to FIG. 8, the venturi **75** may include a primary water input **165** which may feed water from the cold water input **65** via the fourth water line **70**. The venturi **75** may also include the expansion vessel input **170** which may feed water from the expansion vessel **50**. The venturi **75** may include a water outlet **40** where the water mixed from the cold water input **65** and expansion vessel **50** may exit the venturi **75**. When installed, such output may flow out of the venturi **75** into the sixth water line **85**. The venturi **75** may include a larger circumference at each end **175** of the venturi **75** and a smaller circumference at a center point **180** of the venturi **75**. The difference in the circumference may allow the water to reach a higher velocity while traveling through the venturi **75**. The low pressure venturi **150** may include a larger center circumference than the high pressure venturi **145** with a smaller center circumference.

[0049] Turning to FIG. 9, the wait water collection system **1** may include similar components as

those described above. However, the expansion vessel **50** may include a separate vessel input **185** and vessel output **190**. In such a scenario, water may flow from the thermal diverter **15** into the expansion vessel **50** via the vessel input **185**, and water may flow out of the expansion vessel **50** into the venturi **75** via the vessel output **190**.

[0050] Turning now to FIG. **10**, the wait water collection system **1** may include additional pressure sensors and/or thermal diverters, which may assist with overcoming certain failures. If the wait water collection system **1** is in normal operation, when a user switches on the mixer valve **20**, a pressure of the second water line **35** may reduce. The pressure sensor **90** may sense such a reduction in pressure. As a result, the inlet gate valve **95** may open, which allows tepid water from the hot water input **5** to flow to the thermal diverter **15** and into the expansion vessel **50**. Additionally, the drain gate valve **100** may close, which may cause the water collected in the expansion vessel **50** to flow to the venturi **75** if there is a cold water element of the user's temperature setting.

[0051] When the shower is switched off before the thermal diverter **15** has received water meeting or exceeding the temperature threshold, such water may travel to the expansion vessel **50**. In such a situation, pressure in the second water line **35** may or may not be able to be re-pressurized. This may cause the inlet gate valve **95** to remain open, allowing water to continue flowing into the expansion vessel **50**, even though the user has switched off the mixer valve **20** and left the shower. This may cause water to fill the expansion vessel **50**. Because the drain gate valve **100** may be closed due to the low pressure in the second water line **35**, the wait water collection system **1** could eventually fill and “lock up.” In this condition, if the mixer valve **20** is switched “on,” no more water can enter the expansion vessel **50** due to being full, and the wait water collection system **1** may not function as described above. To resume the process described above, the mixer valve **20** may be turned “on” and a cold temperature may be selected, which may cause the expansion vessel **50** to empty via the venturi **75**. Then, a user may select a warmer temperature, which allows hot water to reach the thermal diverter **15** and re-pressurize the second water line **35**. However, this process may be time consuming and lead to wasted water.

[0052] The wait water collection system **1** may include a pressure sensor **191** on the fourth water line **70**. The pressure sensor **191** may detect a pressure from the cold water input **65**. The pressure may be sent via a pilot tube **192** to a tepid gate valve **193** on the third water line **45**. The detected pressure may cause the tepid gate valve **193** to open or close. Pressure may be higher when the shower is not flowing, while pressure may be lower when the shower is flowing. Such lower pressures may be as low as 0.1 Bar (1.45 psi), which may be detectable by the pressure sensor **191**. When the pressure is higher, the tepid gate valve **193** may close, preventing water from entering the expansion vessel **50**. When the pressure is lower, the tepid gate valve **193** may open, causing water to flow into the expansion vessel **50**. This may aid in ensuring that water stops flowing to the expansion vessel **50** when the mixer valve **20** is switched “off.”

[0053] A thermostatic diverter **195**, as illustrated in FIG. **11**, may be used to divert water to and from different outlets based on temperature of media entering the diverter **195**. The thermostatic diverter **195** may be used in place of the thermal diverter **15** as discussed above with reference to FIGS. **1-10**. The thermostatic diverter **195** may include a supply feed **200**. The supply feed **200** may be a supply of any media (e.g., hot water). The water from the supply feed **200** may enter the thermostatic diverter **195** and may be diverted to a first outlet **205** and/or a second outlet **210**. The thermostatic diverter **195** may direct the media to only one of the first outlet **205** or the second outlet **210**, or to both of the first outlet **205** and the second outlet **210** simultaneously. The first outlet **205** may be coupled or connected to a storage tank. As one example, the storage tank may store cold water. The second outlet **210** may be coupled or connected to a faucet (e.g., shower, sink faucet, etc.). The thermostatic diverter **195** may cause water that reaches a certain or desired temperature to exit the diverter **195** via the second outlet **210**, while water that has not reached such certain temperature is directed to exit the diverter **195** via the first outlet **205**.

[0054] Turning to FIG. 12, a diaphragm 215 may control flow into and through the thermostatic diverter 195. As such, when the diaphragm 215 is in a closed position due to a valve (e.g., shower valve, faucet valve, etc.) connected to the diverter 195 being closed, no fluid (e.g., water) may flow through the thermostatic diverter 195. As illustrated in FIG. 12, the diaphragm 215 is in a closed position. The thermostatic diverter 195 may be activated when flow is initiated by opening the valve connected to the thermostatic diverter 195. When flow is initiated, pressure at a first end 220 of a check valve 225 may drop, which may cause a diaphragm 215 to adjust to an open position due to a pressure imbalance. When in a closed position due to the shower valve, faucet valve, etc. [0055] being closed, the check valve 225 may trap pressure between the check valve 225 and the diaphragm 215, which may close the diaphragm 215. When the diaphragm 215 is closed, water may be prevented from flowing into or out of the thermostatic diverter 195.

[0056] When the diverter 195 is activated via the diaphragm 215 opening, water, which may flow into the diverter 195 via the supply feed 200, may pass over a thermostat 230. When water has not reached a certain temperature, the thermostat 230 may bias a shuttle 235 to direct water to the first outlet 205, which may feed, for example, into a storage tank. When water has reached a certain temperature, the thermostat 230 may bias a shuttle 235 to direct water to the second outlet 210. The thermostat 230 may extend to bias the shuttle 235 in such a direction. As one example, the temperature at which the water should be diverted to the second outlet 210 is pre-set or pre-determined. As another example, an adjustment system 240 may be used to select a temperature at which the water should be diverted from the first outlet 205 to the second outlet 210. A user may input a desired temperature into the adjustment system 240 (e.g., through a user interface), which may be coupled to control translation of the thermostat 230.

[0057] Turning now to FIG. 13, water may enter a bathroom segment 245 of a wait water collection system via an input line 250. The entire wait water collection system may include a combination of the components described herein. The water may be cold water from a municipal water supply. The input line 250 may feed water to a shower 255, storage tank 260, and/or any other desirable outlet. The input line 250 may feed water through a heater 265 to be heated. While being heated or while stored in a diverter input line 270, the water may not necessarily be a desired temperature. For example, after a shower is shut off, water may sit within the diverter input line 270 and cool off. To determine whether the water has reached said desired temperature, the water may be fed into a thermostatic diverter 275 via the diverter input line 270. The thermostatic diverter 275 may be similar to or the same as the thermostatic diverter 195 as discussed with reference to FIGS. 11 and 12.

[0058] If the water has reached the desired temperature, the water may travel to the shower 255 via a hot water line 280. If the water has not reached the desired temperature, the water may be stored in the storage tank 260 for later or other use. The water in the storage tank 260 may provide water to a toilet 285 or to any other desirable outlet (e.g., a cold water faucet). After used in the shower 255, toilet 285, or other outlet, the water may flow to a septic system via septic lines 290.

[0059] Instead of to the storage tank 260 before being directed to the toilet 285, as illustrated in FIG. 14, water may be directly fed into a toilet 295 from the thermostatic diverter 275. In such configuration, the toilet 295 may include a toilet storage tank 300 and a flush tank 305, as illustrated in FIG. 15. The toilet storage tank 300 may sit on top of the flush tank 305, may be placed in a wall, or may take any other suitable configuration now known or hereafter developed. The toilet 295 may include a diversion valve which may be used to adjust whether water from the toilet storage tank 300 or the flush tank 305 is used in a toilet bowl 310.

[0060] As illustrated in FIG. 16, hot water may be stored in a highly insulated hot water storage tank 315. Water may be directed from a supply input 320 to a heater 325 in order to be heated. The water may be fed to the storage tank 315 via a storage input line 330. Even if the water from the heater 325 is not sufficiently heated, the water may mix with the hot water in the storage tank 315 in order to be adequately heated to a desired temperature. Water from the input 320 and from the

storage tank **315** may be fed to a shower **335**. After use in the shower **335**, water may flow to a septic via a septic line **340**.

[0061] Turning now to FIG. **17**, hot water may enter a wait water collection system **345** via a hot water input line **350**, and cold water may enter the wait water collection system **345** via a cold water input line **355**. The cold water may be directed directly to a shower mixer valve **360**. The hot water may be fed into a thermostatic diverter **365**, which may be similar to or the same as the thermostatic diverter **395** of FIGS. **11** and **12**. When a valve associated with a shower outlet **370** is opened, pressure (via a pressure feedback loop **375**) may cause a check valve **380** of the diverter **365** to adjust. A diverter switch **385** may turn “on” and cause water to flow into the diverter **365**.

[0062] If water has not reached a sufficient or desired temperature, then the water may be fed into a reservoir **390**. From the reservoir **390**, water may be directed to a check valve **395** and into a venturi **400**. The venturi **400** may be used to introduce a slow flow rate throughout the use of the shower. If the water has reached the desired temperature, then the water may be fed into the shower mixer valve **360**. The shower mixer valve **360** may mix the cold water and the hot water and may direct the mixed water to the venturi **400**. From the venturi **400**, water may exit the wait water collection system **345** via the shower outlet **370**. Such a configuration may be used as a mechanical mixer.

[0063] As illustrated in FIG. **18**, hot water and cold water may be directed into the shower mixer valve **405** of a wait water collection system **410** via a hot water input line **415** and a cold water input line **420**, respectively. The mixed water may then be directed to a thermostatic diverter **425**, which may be similar to or the same as the thermostatic diverter **195** of FIGS. **11** and **12**. The diverter **425** may determine whether the mixed water has reached a desired temperature.

[0064] If the mixed water has not reached the desired temperature, then the insufficiently heated water may be directed to a reservoir **430**. The water may then be directed to a check valve **435** and to a venturi **440**. The venturi **440** may be used to introduce a slow flow rate throughout the use of the shower. If the mixed water has reached the desired temperature, then the water may be directed from the mixer valve **425** to the venturi **440**. The water from the reservoir **430** and/or the thermostatic diverter **425** may be fed to a shower outlet **445** via the venturi **440**. Such a configuration may be used as a mechanical mixer.

[0065] Turning to FIG. **19**, hot water may be inputted into a wait water collection system **450** via a hot water input line **455**. The hot water may be fed into a thermostatic diverter **460**, which may be similar to or the same as the thermostatic diverter **195** of FIGS. **11** and **12**. When a valve associated with a shower outlet **465** is opened, pressure (via a pressure feedback loop **470**) may cause a check valve **475** of the diverter **460** to adjust. A diverter switch **480** may turn “on” and cause water to flow into the diverter **460**. If water has not reached a sufficient or desired temperature, then the water may be fed into a reservoir **485**. From the reservoir **485**, water may be directed to a pump **490** and then to a check valve **495**.

[0066] If the water has reached the desired temperature, then the water may be fed into a shower mixer valve **500** from the diverter **460**. The shower mixer valve **400** may mix the hot water from the thermostatic diverter **460** and cold water, which may be fed into the system **450** via a cold water input line **505** passing through a check valve **510**. Water may exit the wait water collection system **450** via the shower outlet **465**. Such a configuration may be used as a mechanical mixer with a preferred reintroduction point using a pump.

[0067] As illustrated in FIG. **20**, hot water and cold water may be inputted into a wait water collection system **515** via a hot water input line **520** and a cold water input line **525**, respectively. The system **515** may include a hot water pressure limiter **530** along the hot water input line **520** prior to a diverter switch **535**. The system **515** may include a cold water pressure limiter **540** along the cold water input line **525** between a check valve **545** and a venturi **550**. The hot water pressure limiter **530** and cold water pressure limiter **540** may be any pressure limiter known in the art, including a Neoperl® pressure limiter. The hot water pressure limiter **530** and cold water pressure

limiter **540** may adjust the pressure of the hot water in the hot water input line **520** or cold water in the cold water input line **525**, respectively, to a pressure set point, which may be selected by a user. The venturi **550** may be used to introduce a slow flow rate throughout the use of the shower. [0068] The hot water may be fed from the hot water pressure limiter **530** into a thermostatic diverter **555**, which may be similar to or the same as the thermostatic diverter **195** of FIGS. **11** and **12**. When a valve associated with a shower outlet **560** is opened, pressure (via a pressure feedback loop **565**) may cause a check valve **570** of the diverter **555** to adjust. The diverter switch **535** may turn “on” and cause water to flow into the diverter **555**. If the water has not reached a sufficient or desired temperature, then the water may be fed into a reservoir **575**. From the reservoir **575**, water may be directed to a check valve **580**, to the venturi **550**, and to a shower mixer valve **585**. If the water has reached the desired temperature, then the water may be fed into the shower mixer valve **585** from the check valve **570**. The shower mixer valve **585** may mix the hot water and cold water. Water may exit the wait water collection system **515** via the shower outlet **560**. Such a configuration may be used as a mechanical mixer with a preferred reintroduction point using a venturi.

[0069] As illustrated in FIG. **21**, a wait water collection system **590** may utilize a digital mixer scheme. Hot water may be fed into the system **590** from a hot water input line **595**. The hot water may be fed through a temperature sensor **600** and into a digital mixer valve dual outlet **605**, and cold water may be fed directly into the digital mixer valve **605**. The temperature sensor **600** may be coupled to the digital mixer valve **605** to determine whether the water passing into the digital mixer valve **605** has reached a certain temperature. If so, then the water may flow through a venturi **610** and a temperature sensor **615** to exit via a shower outlet **620**. The temperature sensor **615** may provide feedback to the digital mixer valve **605** regarding temperature of the water passing out of the digital mixer valve **605**. If not, then the water may flow into a reservoir **625**, which may then flow through a check valve **630** into the venturi **610** to exit via the shower outlet **620**.

[0070] Turning now to FIG. **22**, a wait water collection system **635** may utilize a digital mixer scheme with a preferred reintroduction point. Cold water may be fed through a check valve **640** and into a digital mixer valve **645**, and hot water may be fed through a temperature sensor **650** and into the digital mixer valve **645**. The temperature sensor **650** may be coupled to the digital mixer valve **645** to determine whether the water passing through the digital mixer valve **645** has reached a certain temperature. If so, then the water may flow out via shower outlets **655**. If not, then the water may flow into a reservoir **660** for storage. If needed, the water may flow from the reservoir **660** through a check valve **665** and may be reintroduced to the digital mixer valve **645** via a pump **670**.

[0071] As illustrated in FIG. **23**, a wait water collection system **675** may include a hot water pressure limiter **680** along a hot water input line **685**. The hot water input line **685** may input hot water into the system **675**, and the hot water may pass through a temperature sensor **690** after passing through the hot water pressure limiter **680**. Cold water may be introduced into the system **675** via a cold water input line **695**. The system **675** may include a cold water pressure limiter **700** along the cold water input line **695** between a check valve **705** and a venturi **710**. The hot water pressure limiter **680** and cold water pressure limiter **700** may be any pressure limiter known in the art, including a Neoperl® pressure limiter. The hot water pressure limiter **680** and cold water pressure limiter **700** may adjust the pressure of the hot water in the hot water input line **685** or cold water in the cold water input line **695**, respectively, to a pressure set point, which may be selected by a user.

[0072] The cold water and the hot water may be fed into a digital mixer valve **715** from the venturi **710** and the temperature sensor **690**, respectively. The venturi **710** may be used to introduce a slow flow rate throughout the use of the shower. The temperature sensor **690** may be coupled to the digital mixer valve **715** to provide an indication as to whether the water passing through the digital mixer valve **715** has reached a certain temperature. If so, then the water may flow out via shower outlets **720**. If not, then the water may flow into a reservoir **725** for storage. If needed, the water

may flow from the reservoir **725** through a check valve **730** and may be reintroduced to the digital mixer valve **715** via the venturi **710**. Such a configuration may be used as a digital mixer with a preferred reintroduction point using a venturi.

[0073] Turning now to FIG. **24**, a water supply **735** may provide water via a cold water line **740** to a water heater **745**, a washer **750**, a plurality of sinks **755**, and/or a shower **760**. The water heater **745** may heat the water from the water supply **735** and may provide heated water via a hot water line **765** to the washer **750**, the plurality of sinks **755**, and/or the shower **760**. Water may be stored in accumulator tanks **770** associated with the sinks **755** and/or the shower **760**. The accumulator tanks **770** may be used to store the water until a desired temperature of the water is reached, and the stored water, which may not have reached the desired temperature, may be used as an input to the cold water line **740**.

[0074] As illustrated in FIG. **25**, a float valve **775** may be used within a reservoir **780** to determine whether the reservoir **780** is full. If the float valve **775** indicates that the reservoir **780** is full, then water may be directed away from the reservoir **780**. The reservoir **780** may be positioned a certain distance **785** between both of a hot water outlet **790** and a cold water outlet **795**, and such distance **785** may be associated with the static head pressure.

[0075] Turning now to FIG. **26**, a water management pipe **800** may provide water from a source to a user. The pipe **800** may be substantially cylindrical in shape and may include a first end **805** and a second end **810** opposing the first end **805**. The first end **805** may receive, be received by, or otherwise be coupled to an external source (not shown) of water or other fluid, which will collectively be referred to hereinafter as “water.” Alternatively, the first end **805** may be attached to separate piping, and that separate piping may be connected to an external water source, such that the pipe **800** is positioned downstream of the water source. The second end **810** may provide water to or toward a user. Alternatively, the second end **810** may receive, be received by, or otherwise be coupled to an appliance, attachment, or other mechanism (not shown) for providing water. For example, the second end **810** may be in fluid communication with a showerhead, faucet, or any other suitable means for supplying water to a user. As with the first end **805**, the second end **810** may be attached to separate piping, and that separate piping may be deliver water to a user or to another end use, such that pipe **800** is positioned upstream of the user or end use. Thus, water may enter the pipe **800** at the first end **805**, travel through the pipe **800** along a path that is substantially aligned with a central longitudinal axis A of the pipe **800**, and exit the pipe **800** at the second end **810** where it may be provided to a user. The pipe **800** may include bends or curves.

[0076] The pipe **800** may include a shell **815** and a flexible membrane **820** positioned within the shell **815**. The shell **815** may be substantially cylindrical in shape, or another shape as desired. The shape and structure of the shell **815** may be fixed, whereas the shape and structure of the flexible membrane **820** may vary. In general, a cross section of the flexible membrane **820** taken in a direction substantially perpendicular to the axis A at any point along the pipe **800** may have a generally circular profile, or any other fixed profile as desired. However, the shape and structure of the flexible membrane **820** may vary depending on the flow of water through the pipe **800**.

Additionally, the shell **815** and the flexible membrane **820** may be provided as coaxial structures. For example, the shell **815** and flexible membrane **820** may each be formed around and extend along the same central axis A.

[0077] The flexible membrane **820** may be coupled to the shell **815** at the first end **805** and the second end **810** of the pipe **800**, or anywhere in between. Thus, while the shape and structure of the flexible membrane **820** between the first and second ends **805**, **810** may change during various stages of operation of the pipe **800**, the flexible membrane **820** may be securely positioned within the shell **815**. The flexible membrane **820** may be coupled to the shell **815** such that, when the pipe **800** is in use, water may flow through the space defined by the flexible membrane **820**. For example, when the pipe **800** is in use, water may be excluded from a space located between the flexible membrane **820** and the shell **815**. Instead, water may be restricted to flowing through the

flexible membrane **820**.

[0078] Turning to FIGS. **27** and **28**, the shape and structure of the flexible membrane **820** may differ between a first stage of operation and a second stage of operation of the pipe **800**. During the first stage of operation, illustrated in FIG. **27**, water may not be flowing through the pipe **800**. The flexible membrane **820** may occupy an idle position during the first stage of operation. During the second stage of operation, illustrated in FIG. **28**, water may be flowing through the pipe **800**. The flexible membrane **820** may occupy an active position during the second stage of operation.

[0079] The flexible membrane **820** may include or be formed from a phase change material (PCM). The PCM may be provided in the form of a salt hydrate, paraffin wax, non-paraffin organic, metal alloy, or any other suitable material. The PCM may facilitate the flexible membrane **820** transitioning from the idle position to the active position. For example, the PCM may cause or allow the shape and structure of the flexible membrane **820** to change in response to the flow of water through the pipe **800**.

[0080] As illustrated in FIG. **27**, during the first stage of operation, the flexible membrane **820** may occupy an idle position. In the idle position, the flexible membrane **820** may reduce the effective internal diameter of the pipe **800**. For example, the shell **815** may have an internal diameter **D1**. The internal diameter **D1** of the shell **815** may be substantially constant throughout the pipe **800**. Alternatively, the internal diameter **D1** may vary. The flexible membrane **820** may reduce the effective internal diameter of the pipe **800** to an amount less than the internal diameter **D1** of the shell **815**. For example, an internal diameter **D2** of the flexible membrane **820** may be less than the internal diameter **D1** of the shell **815**. Additionally, the internal diameter **D2** may vary at different points along the pipe **800**. At the first and second ends **805**, **810** where the flexible membrane **820** is coupled to the shell **815**, the internal diameter **D2** of the flexible membrane **820** may be substantially equal to or minimally less than the internal diameter **D1** of the shell **815**. Thus, at the first and second ends **805**, **810**, the effective internal diameter of the pipe **800** may be reduced by a minimal amount or not at all.

[0081] However, as the flexible membrane extends away from the first and second ends **805**, **810** and toward a midpoint **825** of the pipe **800**, the internal diameter **D2** of the flexible membrane **820** may decrease. As the internal diameter **D2** decreases, the effective internal diameter of the pipe **800** may be reduced and the flexible membrane may at least partially restrict the flow of water through the pipe **800**. In the example of FIG. **27**, as the flexible membrane **820** approaches the midpoint **825**, the internal diameter **D2** may decrease to essentially zero or close to zero. Thus, as the flexible membrane **820** approaches the midpoint **825**, the effective internal diameter of the pipe **800** may be reduced to essentially zero or close to zero, thereby restricting the rate of flow of water through the pipe **800** to essentially zero or close to zero.

[0082] In addition to the example of FIG. **27**, the internal diameter **D2** of the flexible membrane **820** at the midpoint **825** may have any other value less than the internal diameter **D1** of the shell **815**. Alternatively, the internal diameter **D2** of the flexible membrane **820** at the midpoint **825** may have any other value less than the internal diameter **D2** of the flexible membrane **820** at the first and second ends **805**, **810**.

[0083] The flexible membrane **820** may be imparted with an inherent structural bias toward the idle position illustrated in FIG. **27**. For example, the construction of the flexible membrane **820** may apply an inward pressure that biases the flexible membrane **820** away from the shell **815** and toward the axis **A**. Thus, when no external force or stimulus is applied, the flexible membrane **820** may default to occupying the idle position. When the flexible membrane **820** is in the idle position, the pipe **800** may include dead space **830** positioned between the flexible membrane **820** and the shell **815**. For example, as the internal diameter **D2** of the flexible membrane **820** decreases, a volume of the dead space **830** may increase. Water may be excluded from the dead space **830** or may be restricted from entering the dead space **830**. In this way, the flexible membrane **820** may prevent or restrict the accumulation of stagnant water within the pipe **800** when the pipe **800** is not

in use.

[0084] However, the flexible membrane **820** may change shape in response to external forces or stimuli. For example, when water flows into the pipe **800** via the first end **805**, the flowing water may apply an outward pressure to the flexible membrane **820**. The outward pressure applied by the flowing water may directly or indirectly oppose the inherent structural bias of the flexible membrane **820** toward the idle position. Thus, pressure applied to the flexible membrane **820** by the flowing water may tend to cause the flexible membrane **820** to move from the idle position toward the active position. The pressure may increase the internal diameter D2 of the flexible membrane **820** and may permit water to flow more freely from the first end **805** to the second end **810**.

[0085] Additionally, the PCM of the flexible membrane **820** may contribute to or facilitate the flexible membrane **820** moving from the idle position to the active position. The PCM may cause the flexible membrane **820** to change shape more readily in response to external stimuli. For example, heated water may enter the pipe **800** via the first end **805** and come into contact with the flexible membrane **820**. The PCM of the flexible membrane **820** may absorb thermal energy from the heated water. This thermal energy may, for example, cause the flexible membrane **820** to become more malleable. In this way, the heated water may cause the flexible membrane **820** to be more susceptible to being pushed toward the shell **815** by the pressure of water flowing through the pipe **800**.

[0086] If water ceases flowing through the pipe **800**, the PCM of the flexible membrane **820** may release thermal energy, causing the flexible membrane **820** to become less malleable. Additionally, the cessation of water flowing through the pipe **800** may eliminate or reduce the outward pressure applied to the flexible membrane **820**. As the flexible membrane **820** loses malleability, and without the outward pressure of water flowing through the pipe **800**, the inherent structural bias of the flexible membrane **820** may cause the flexible membrane **820** to return to the idle position.

[0087] Turning to FIG. **28**, the flexible membrane **820** may occupy the active position during the second stage of operation of the pipe **800**. In the active position, the shape of the flexible membrane **820** may be substantially cylindrical and may mirror the shape of the shell **815**. Thus, when the flexible membrane **820** is in the active position, the flexible membrane **820** may contact the shell **815** and the dead space **830** of FIG. **27** may be substantially or entirely eliminated. For example, in the active position, the internal diameter D2 of the flexible membrane **820** may be substantially constant throughout the pipe **800**. Additionally, in the active position, the internal diameter D2 of the flexible membrane **820** may be substantially equal to or minimally less than the internal diameter D1 of the shell **815** throughout the pipe **800**. In the second stage of operation of the pipe **800**, the flexible membrane **820** may be maintained in the active position by the outward pressure applied by the flowing water, by the enhanced malleability or other characteristic of the flexible membrane **820** caused by the PCM, or by a combination thereof. Thus, in the second stage of operation, water may flow freely through the pipe **800**.

[0088] The aforementioned storage tanks and reservoirs may be used to store cold water or hot water. As discussed above, water in a hot water input line may cool off while within the line after a shower is shut off, specifically if the climates surrounding the hot water input line are colder. Similarly, water in a cold water input line may warm up while within the line after a shower is shut off, specifically if the climates surrounding the cold water input line are warmer. It should be understood that the storage tanks and reservoirs, although discussed with reference to storing cold water, may store any temperature of water. Further, in order to aid in preventing bacterial growth mitigation in a water storage tank, certain cleansing of the water may occur prior to entering the tank. As one example, ultraviolet light may be used to cleanse the water.

[0089] As is evident from the foregoing description, certain aspects of the present invention are not limited by the particular details of the examples illustrated herein, and it is therefore contemplated that other modifications, applications, variations, or equivalents thereof, will occur to those skilled

in the art. Many such changes, modifications, variations, and other uses and applications of the present constructions will, however, become apparent to those skilled in the art after considering the specification and the accompanying drawings. In addition, unless mention was made above to the contrary, it should be noted that all of the accompanying drawings are not to scale. All such changes, modifications, variations, and other uses and applications which do not depart from the spirit and scope of the present inventions are deemed to be covered by the inventions which are limited only by the claims which follow.

Claims

1. A wait water collection system comprising: at least one water input coupled to a water line; a thermal diverter positioned on the water line, wherein the thermal diverter adjusts based on whether a temperature of water from the at least one water input meets a temperature threshold; an expansion vessel; a mixer valve; and an outlet; wherein when the temperature of the water meets or exceeds the temperature threshold, the water is directed to the mixer valve; and wherein when the temperature of the water falls below the temperature threshold, the water is directed to the expansion vessel.
2. The wait water collection system of claim 1, further comprising a venturi for drawing water out of the expansion vessel.
3. The wait water collection system of claim 1, further comprising a pressure sensor for determining a pressure within the water line.
4. The wait water collection system of claim 3, further comprising an inlet gate valve and a drain gate valve.
5. The wait water collection system of claim 4, wherein when the pressure sensor determines that the pressure is high, the inlet gate valve closes and the drain gate valve opens.
6. The wait water collection system of claim 4, wherein when the pressure sensor determines that the pressure is low, the inlet gate valve opens and the drain gate valve closes.
7. The wait water collection system of claim 4, wherein when the drain gate valve is open, water from the expansion vessel flows directly out of the wait water collection system via the outlet.
8. The wait water collection system of claim 4, wherein when the drain gate valve is closed, water from the expansion vessel flows to the mixer valve.
9. A wait water collection system comprising: a first media input providing a first media to a first media line; a second media input providing a second media to a second media line; a thermal diverter positioned on the first media line, wherein the thermal diverter adjusts based on whether a temperature of the first media meets a temperature threshold; a pressure sensor positioned on the first media line, wherein the pressure sensor determines a pressure within the first media line; an expansion vessel, wherein when the temperature of the first media falls below the temperature threshold, the thermal diverter causes the first media to flow into the expansion vessel; a venturi positioned on the second media line, wherein when the first media fills the expansion vessel, the venturi draws the first media out of the expansion vessel into the second media line; a mixer valve coupled to each of the first media line and the second media line, wherein the first media and the second media mixes into a water output; and an outlet for providing the water output to a user.
10. The wait water collection system of claim 9, further comprising an inlet gate valve and a drain gate valve.
11. The wait water collection system of claim 10, wherein when the pressure sensor determines that the pressure is high, the inlet gate valve closes and the drain gate valve opens.
12. The wait water collection system of claim 10, wherein when the pressure sensor determines that the pressure is low, the inlet gate valve opens and the drain gate valve closes.
13. The wait water collection system of claim 10, wherein when the drain gate valve is open, the first media from the expansion vessel flows directly out of the wait water collection system via the

outlet.

14. The wait water collection system of claim 10, wherein when the drain gate valve is closed, the first media from the expansion vessel flows to the mixer valve via the second media line.

15. The wait water collection system of claim 9, wherein when the temperature of the first media meets or exceeds the temperature threshold, the first media is directed to the mixer valve.

16. The wait water collection system of claim 9, wherein the first media is hot water, and wherein the second media is cold water.

17. A wait water collection system comprising: at least two water inputs coupled to at least two water lines; a thermal diverter positioned on one of the at least two water lines, wherein the thermal diverter adjusts based on whether a temperature of water from the one of the at least two water inputs meets a temperature threshold; and an expansion vessel; wherein when the temperature of the water falls below the temperature threshold, the water is directed to the expansion vessel.

18. The wait water collection system of claim 17, wherein the expansion vessel has a capacity of 6 to 16 liters.

19. The wait water collection system of claim 17, wherein the expansion vessel includes an expansion vessel port for at least one of inputting and outputting water into the expansion vessel.

20. The wait water collection system of claim 17, wherein when a pressure in the wait water collection system is high, a drain gate valve is opened, and water from the expansion vessel flows out of the wait water collection system via an outlet.
